

What triggered this

The panel below reports the test-loss chord barrier (the one we’ve used so far) after selecting the best overall alignment separately for every ordered stage pair. At the early checkpoints, aligning A_{200} with B_n gives a much smaller barrier than aligning A_n with B_{200} . For example, at $n = 0$ the means are 0.300 and 0.699, respectively; at $n = 1$ they are 0.316 and 0.589.

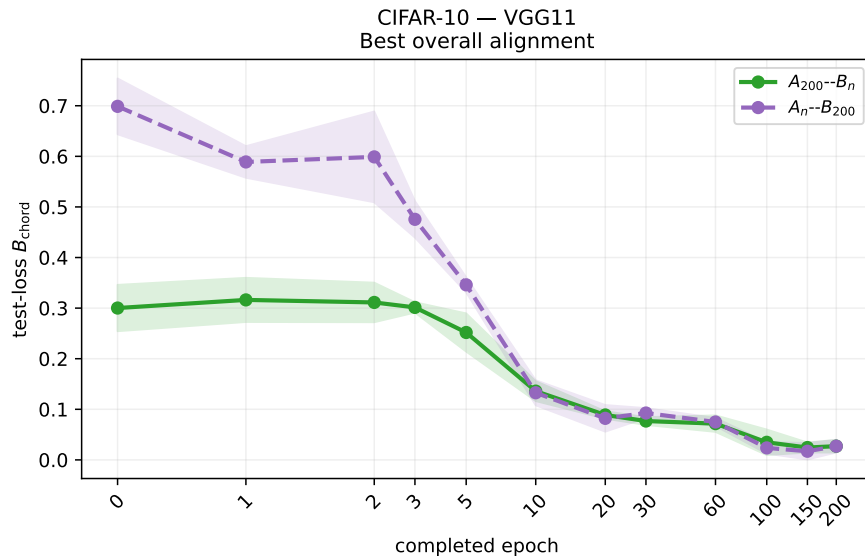


Figure 1: Mean $\pm$ one standard deviation over three seed pairs.

This is not just an $A-B$ path traversed backwards. The two comparisons are $A_{200}-B_n$ and A_n-B_{200} , so they use different checkpoints. They should nevertheless have the same distribution if the A/B roles and the alignment procedure are exchangeable.

At $n = 0, 1, 2, 3$, both VGG11 directions selected Sinkhorn followed by scale fine-tuning. Across the full 12×12 ordered matrix, Sinkhorn+scale was selected in 91 cells and almost always when n was small (big difference in training stages between networks), compared with 53 cells for WM+scale (performing better when n was big, indicating small difference in networks’ training stages) .

Fixed-method check: CIFAR-10 / VGG11

To separate method selection from method behaviour, the next two plots keep the method fixed. “Sinkhorn with scale” here means a hard Sinkhorn permutation followed by positive-scale fine-tuning. The second method is WM+scale.

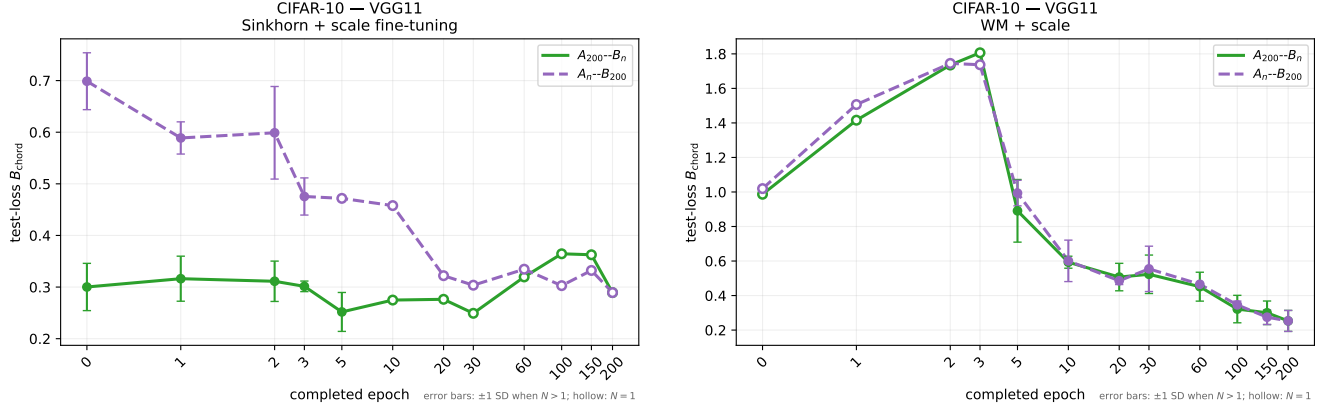


Figure 2: VGG11 fixed-method diagnostic. Sinkhorn+scale is strongly directional for early/final pairs. WM+scale is much closer to symmetric, especially at the earliest stages. Error bars show ± 1 SD wherever multiple seed pairs were evaluated; hollow points have only one pair.

Short interpretation

The permutation-matching part of WM+scale approximately solves

$$P^* = \arg \min_P \|\theta_A - P\theta_B\|_2^2.$$

For an exact solution on the same endpoint pair, swapping A and B simply uses the inverse permutation, so the objective itself is direction-independent. The implementation is greedy and the plotted stage slices are not literal reversals, so small WM+scale differences are still possible; however, as seen in the plot, the directional variation is very small.

The Sinkhorn implementation is operationally directional. One endpoint is kept fixed, while a permutation (and then scales) is optimized only for the other endpoint using interpolation loss. The transform starts at identity; the relevant difference is which checkpoint is fixed and which checkpoint is allowed to move. Empirically, in the early/final comparison,

early model $\longrightarrow$ fixed final model

works substantially better than

final model $\longrightarrow$ fixed early model.

I observed similar results for the MLP trained on Fashion-MNIST.